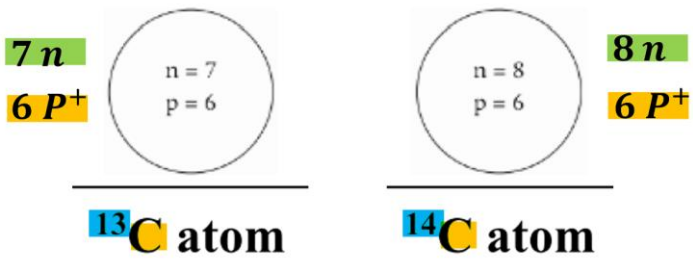
Chapter 5 – Atomic Structure

Isotope

Hydrogen	H	$1\,P^+$	$1e^-$	$0\,n$
	H	$1\,P^+$	$1e^-$	$1\,n$
	H	$1\,P^+$	$1e^-$	$2\,n$
	H	$1\,P^+$	$1e^-$	$100n$

Isotopes are atoms of the element with **same number of protons** but **different numbers of neutrons.**

Isotopes have **SAME chemical properties** because they are atoms with **same electronic arrangement**.



They **CANNOT** be separated by chemical methods. In addition, isotopes have **slightly different physical properties** (e.g. mass). Therefore, they CAN be separated by physical method.

Note: Protons and electrons exist in ALL atoms, but not neutrons (e.g. ^1_1H atom does not have neutrons)

Structure Questions

Q1. Deuterium (^2H) is an isotope of hydrogen. Deuterium reacts with chlorine-35 (^{35}Cl) to form a compound R.

(a) What is the mass number of deuterium?

(1 mark)

(b) Compound S formed from the reaction of chlorine-35 with another isotope of hydrogen protium (^1H). Do compounds R and S have the same chemical properties? Explain your answer.

(1 mark)

Q2. Nihonium (Nh) is a metal element discovered by a team of Japanese scientists in 2004. It can be synthesized by the bombardment of bismuth with zinc.

$$^{209}_{83}Bi + ^{70}_{30}Zn \rightarrow ^{278}_{113}Nh + ^{1}_{0}n$$

(a) (i) Which types of subatomic particle does $^{1}_{0}n$ represent?

(1 mark)

(ii) Explain whether $^{1}_{0}n$ will be deflected by an electric field.

(1 mark)

(b) Nihonium has several isotopes.

(i) What is meant by the term 'isotopes'?

(1 mark)

Q3. Fluorine-19 is the only isotope of fluorine.

(a) (i) Draw the electron diagram of a fluorine atom.

(1 mark)

(b) Describe the structure of a fluorine atom. Your answer should include the type and number of the subatomic particles present in the atom, and the location and the charge of each type of these subatomic particles.

(3 marks)

Relative atomic mass

Relative atomic mass is the **weighted average** of the relative isotopic masses of all of its naturally occurring isotopes of the ^{12}C = 12.00 scale

Example 1

If the abundance of ^{35}Cl and ^{37}Cl are 75% and 25% respectively, calculate the relative atomic mass of Cl.

Relative atomic mass = ^{35}Cl **x a%** + ^{37}Cl **x b%**

Relative isotopic mass of the ^{35}**Cl** isotope = 35

Relative isotopic mass of the ^{37}**Cl** isotope = 37

Solution: Relative atomic mass =

Example 2

Magnesium has 3 isotopes, ^{24}Mg, ^{25}Mg and ^{26}Mg. If the abundance of ^{24}Mg is 78.6% and the relative atomic mass of Mg is 24.327, calculate the abundance of ^{25}Mg.

Relative atomic mass = ^{24}Mg **x a%** + ^{25}Mg **x b%** + ^{26}Mg **x c%**

Solution: **Hints : Let x% be the abundance of ^{25}Mg.**

24.327 = (24)(78.6%) + (25)(x%) + (26) (_____)

24.327 =

Example 3

Bromine has 2 isotopes, ^{79}Br and ^{x}Br. If the abundance of ^{79}Br is 55% and the relative atomic mass of Br is 79.9, calculate the isotopic mass of ^{x}Br.

Relative atomic mass = ^{79}Br **x a%** + ^{x}Br **x b%**

Solution:

Answer Keys

Q1 (a) 2

(b) Yes. This is because the isotopes of an element have the same chemical properties.

Q2 (a) (i) Neutron

(ii) No as it is electrically neutral

(b) Different atoms of an element with same no. of protons but different no, of neutrons

Q3 (a) (i)

(b) A fluorine atom has 9 protons, 9 electrons and 10 neutrons.

The protons (positively charged) and neutrons (neutral) are concentrated in the nucleus of fluorine atom.

Electrons (negatively charged) are moving in electron shells surrounding the nucleus of fluorine atom at a very high speed.

Electronic arrangement of fluorine atom: 2, 7

Example 1

If the abundance of ^{35}Cl and ^{37}Cl are 75% and 25% respectively, calculate the relative atomic mass of Cl.

Relative atomic mass = ^{35}Cl **x a%** + ^{37}Cl **x b%**

Solution: Relative atomic mass = (35) (75%) + (37) (25%)

= 35.5 (No unit!)

Example 2

Magnesium has 3 isotopes, ^{24}Mg, ^{25}Mg and ^{26}Mg. If the abundance of ^{24}Mg is 78.6% and the relative atomic mass of Mg is 24.327, calculate the abundance of ^{25}Mg.

Relative atomic mass = ^{24}Mg **x a%** + ^{25}Mg **x b%** + ^{26}Mg **x c%**

Solution: Let x% be the abundance of ^{25}Mg.

24.327 = (24)(78.6%) + (25)(x%) + (26) (100% – 78.6% – x%)

24.327 = 18.864 + (25)(x%) + 5.564 – (26)(x%)

x = 10.1

The abundance of ^{25}Mg is 10.1%

Example 3

Bromine has 2 isotopes, ^{79}Br and ^{x}Br. If the abundance of ^{79}Br is 55% and the relative atomic mass of Br is 79.9, calculate the isotopic mass of ^{x}Br.

Relative atomic mass = ^{79}Br **x a%** + ^{x}Br **x b%**

Solution: 79.9 = (79) (55%) + (x) (100% – 55%)

x = 81 **(Must be integer)**

PERIODIC TABLE 週期表

GROUP 族

atomic number 原子序

relative atomic mass 相對原子質量

I	II											III	IV	V	VI	VII	0
				1 **H** 1.0													2 **He** 4.0
3 **Li** 6.9	4 **Be** 9.0											5 **B** 10.8	6 **C** 12.0	7 **N** 14.0	8 **O** 16.0	9 **F** 19.0	10 **Ne** 20.2
11 **Na** 23.0	12 **Mg** 24.3											13 **Al** 27.0	14 **Si** 28.1	15 **P** 31.0	16 **S** 32.1	17 **Cl** 35.5	18 **Ar** 40.0
19 **K** 39.1	20 **Ca** 40.1	21 **Sc** 45.0	22 **Ti** 47.9	23 **V** 50.9	24 **Cr** 52.0	25 **Mn** 54.9	26 **Fe** 55.8	27 **Co** 58.9	28 **Ni** 58.7	29 **Cu** 63.5	30 **Zn** 65.4	31 **Ga** 69.7	32 **Ge** 72.6	33 **As** 74.9	34 **Se** 79.0	35 **Br** 79.9	36 **Kr** 83.8
37 **Rb** 85.5	38 **Sr** 87.6	39 **Y** 88.9	40 **Zr** 91.2	41 **Nb** 92.9	42 **Mo** 95.9	43 **Tc** (98)	44 **Ru** 101.1	45 **Rh** 102.9	46 **Pd** 106.4	47 **Ag** 107.9	48 **Cd** 112.4	49 **In** 114.8	50 **Sn** 118.7	51 **Sb** 121.8	52 **Te** 127.6	53 **I** 126.9	54 **Xe** 131.3
55 **Cs** 132.9	56 **Ba** 137.3	57 * **La** 138.9	72 **Hf** 178.5	73 **Ta** 180.9	74 **W** 183.9	75 **Re** 186.2	76 **Os** 190.2	77 **Ir** 192.2	78 **Pt** 195.1	79 **Au** 197.0	80 **Hg** 200.6	81 **Tl** 204.4	82 **Pb** 207.2	83 **Bi** 209.0	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
87 **Fr** (223)	88 **Ra** (226)	89 ** **Ac** (227)	104 **Rf** (261)	105 **Db** (262)													

*	58 **Ce** 140.1	59 **Pr** 140.9	60 **Nd** 144.2	61 **Pm** (145)	62 **Sm** 150.4	63 **Eu** 152.0	64 **Gd** 157.3	65 **Tb** 158.9	66 **Dy** 162.5	67 **Ho** 164.9	68 **Er** 167.3	69 **Tm** 168.9	70 **Yb** 173.0	71 **Lu** 175.0
**	90 **Th** 232.0	91 **Pa** (231)	92 **U** 238.0	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)	103 **Lr** (260)